

Software Testing & Automation Guide

Practical Solutions, Workflows, Tool Comparisons & Real-World Scenarios

1. REPETITIVE ACTIONS FOR AUTOMATION

1. Price Tracking & Stock Notifications (e.g., Flipkart / Amazon)

Constantly searching for an item, applying filters, and checking whether the price has dropped or if it's back in stock.

Why automation helps: A script can query product APIs or scrape the page at set intervals and send instant notifications when thresholds are met, saving manual checking time.

2. Daily Check-ins & Claiming Rewards (e.g., Zomato / Flipkart)

Logging into apps daily to click a "Claim Daily Coins" banner or spin a reward wheel.

Why automation helps: An automated tool schedules these API requests or UI taps daily, ensuring rewards are collected consistently without missing streaks.

3. Bulk Unfollowing or Clearing Saved Posts (e.g., Instagram)

Manually clicking through accounts or saved items one-by-one to remove or unsave them.

Why automation helps: A script loops through element selectors and triggers "unsave" or "unfollow" commands continuously in seconds rather than minutes of manual tapping.

2. SELENIUM IDE AUTOMATED LOGIN TEST SETUP

- 1. Install & Open Extension:** Install the *Selenium IDE* browser extension (Chrome/Firefox). Open it and select *"Record a new test in a new project"*.
- 2. Set Base URL & Start Recording:** Name your project (e.g., `OrangeHRM_Test`) and set the Base URL to `https://opensource-demo.orangehrmlive.com/`. Click **Start Recording**.
- 3. Perform Login Steps:** In the target browser, type `Admin` in the Username field, `admin123` in the Password field, and click **Login**.
- 4. Stop & Save:** Return to the Selenium IDE window, click the red **Stop Recording** button, and name your test case (e.g., `ValidLoginTest`).
- 5. Playback Verification:** Click **Play current test case**. The IDE will execute all steps automatically and highlight them in green upon success.

3. KATALON STUDIO UI NAVIGATION

- **Creating a New Test Case:** Navigate to *Tests Explorer* on the left panel → Right-click *Test Cases* folder → Select *New* → *Test Case*.
- **Adding Test Steps:** Open the test case. Use the *Manual View* tab to click *Add* → *Web UI Keyword* (e.g., `Open Browser` , `Set Text` , `Click`), or switch to *Script View* to write Groovy code.
- **Running Tests:** On the top toolbar, click the *Run* drop-down arrow next to the play button, choose the target browser (e.g., *Chrome*), and track logs in the *Log Viewer* at the bottom.

4. AUTOMATION TOOLS COMPARISON TABLE

Tool Name	Type of Testing	Scripting Required	Example Use Case
Selenium	Web Browser Automation (Functional, Regression)	High (Java, Python, C#, JS)	Cross-browser automated regression testing for large web applications.
Cypress	End-to-End (E2E), Integration, Component Testing	Moderate to High (JavaScript / TypeScript)	Real-time frontend web component testing during developer build pipelines.
Katalon Studio	All-in-One (Web, API, Mobile, Desktop)	Low to Moderate (Low-Code GUI or Groovy/Java)	Enterprise end-to-end testing across web and mobile platforms with hybrid teams.
Postman	API Testing (REST, SOAP, GraphQL), Integration	Low (JavaScript for assertions/scripts)	Validating backend REST API responses, status codes, and JSON payloads.

5. REAL-WORLD SCENARIO AUTOMATION

To automate searching for a product on Flipkart, I would use **Cypress** or **Selenium WebDriver**. The automated script launches the browser, navigates to `https://www.flipkart.com` , and handles any initial popup modals by clicking the close button locator. It locates the search bar input field using its CSS selector (`input[name='q']`), types a target keyword such as *"wireless headphones"*, and simulates an `ENTER` key press. The script then applies explicit waits for the results page to load, asserts that the result container exists, applies a price filter using checkbox locators, and extracts the titles and prices of the top three items to display in the test logs.